Quillen Homology For Jordan Algebras

1 Introduction

This draft studies Jordan algebra homology through derived abelianization. The first goal is to identify the Jordan analogue of the cotangent complex before committing to a concrete cochain complex.

2 Jordan Modules

The first technical problem is to identify abelian group objects over a Jordan algebra with the correct module notion. This section will be synchronized with the claim files on Beck modules and square-zero extensions.

3 The Cotangent Complex

Given a morphism of Jordan algebras, the intended cotangent complex is the derived abelianization object in the corresponding slice category.

4 Low-Degree Terms

The expected low-degree cohomology groups recover derivations and square-zero extensions. These interpretations remain conjectural until the module category is fixed.

5 Comparison Results

This section will compare the intrinsic Quillen-style construction with classical Jordan cohomology and the Tits-Kantor-Koecher Lie-theoretic route.

6 Examples

The first examples are zero multiplication algebras, one-dimensional unital algebras, square-zero extensions, and small special Jordan algebras.

References

[Qui70] Daniel G. Quillen. On the (co-)homology of commutative rings. In Michael Barr and Jonathan Beck, editors, *Applications of Categorical Algebra*, volume 17 of *Proceedings of Symposia in Pure Mathematics*, pages 65–87. American Mathematical Society, Providence, RI, 1970.